
Real-Time Adaptive Passenger Flow

Prediction: A Hybrid Model Approach

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By
DIAR BEGISBAYEV

Under the Supervision of
AIVAR SAKHIPOV



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
ASTANA IT UNIVERSITY LLP
KAZAKHSTAN
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Abstract

Accurate real-time prediction of passenger flows in urban bus networks is essential for operational planning, crowd management, and strategic capacity allocation. Existing approaches often fail to jointly capture the complex spatial dependencies between stations, the multi-timescale temporal dynamics of ridership, and the uncertainty inherent in count-based demand data. We present DTS-GSSF (Dual-Timescale Graph Gated Forecasting), a hybrid deep learning architecture designed for multi-horizon passenger flow prediction on the Astana bus network. The model integrates a Gated Recurrent Encoder for temporal feature extraction, adaptive graph propagation for spatial diffusion across the transit network, and multi-head temporal attention for capturing long-range dependencies. LoRA-based adaptation enables efficient fine-tuning for route-level specialization. We evaluate DTS-GSSF on a synthetic dataset of 1.3 million ridership records across 374 bus stations over 12 months, generated from OpenStreetMap topology and realistic demand simulation. On the held-out test set, DTS-GSSF achieves an R-squared of 0.889 (88.9%) and a mean absolute error of 2.43 passengers per station-hour, outperforming baseline approaches including Historical Average, LSTM, GRU, and TCN. Ablation studies confirm that the TemporalAttention mechanism provides the largest architectural improvement (+1.3% R-squared), while the Negative Binomial likelihood appropriately models overdispersed count data. The model converges within 90 epochs using mixed-precision training on consumer-grade hardware, demonstrating practical applicability for real-time transit intelligence platforms.

Keywords: passenger flow prediction, graph neural networks, gated recurrent models, spatiotemporal forecasting, urban transit, deep learning, Astana bus network

Dedication and Acknowledgements

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Finally, I thank my colleagues and friends for their encouragement and for the many stimulating discussions that helped refine the ideas presented in this thesis.

Student's Declaration

I, DIAR BEGISBAYEV, certify that the work embodied in this Master's thesis, carried out by me under the supervision of AIVAR SAKHIPOV for the period of September 2024 to June 2026 at Astana IT University, has not been submitted for the award of any other degree or diploma except where due acknowledgement has been made in the text.

I further declare that the material obtained from other sources has been duly acknowledged in the thesis, and that the research was conducted in accordance with the ethical guidelines of Astana IT University.

SIGNED: DATE:

Certificate by Supervisor

This is to certify that the research work embodied in this thesis entitled “Real-Time Adaptive Passenger Flow Prediction: A Hybrid Model Approach” submitted to Astana IT University, for the award of the degree of Master of Science in Computer Science and Engineering, has been carried out by DIAR BEGISBAYEV under my supervision at Astana IT University, Astana, Kazakhstan, from September 2024 to June 2026.

To the best of my knowledge and belief, this work is original and has not been submitted so far in part or in full for the award of any degree or diploma of any University or Institute.

SIGNED: DATE:

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Chapter 1

Introduction

Urban public transit systems are the backbone of sustainable mobility in rapidly growing cities. In Astana, the capital of Kazakhstan, the bus network serves as the primary mode of public transportation for over one million residents, with demand patterns that fluctuate dramatically across space, time, and weather conditions. Accurate real-time prediction of passenger flows at individual bus stops is critical for operational planning, dynamic vehicle dispatching, and proactive crowd management. Yet accurate prediction is hindered by three compounding factors: the intricate spatio-temporal dependencies among stops, the multi-timescale nature of ridership dynamics, and the inherent uncertainty of human mobility behaviour.

We address the problem of multi-horizon passenger flow prediction on the Astana bus network. Formally, given a historical context window of T hours across N bus stations, we seek to predict the expected number of boarding passengers at each station for the next H_{pred} hours. The input tensor is $\mathbf{x} \in \mathbb{R}^{B \times T \times N \times F}$, where B is the batch size, $T = 72$ hours (a three-day context), $N = 374$ stations, and $F = 16$ engineered features including raw ridership counts, weather variables, calendar indicators, and cyclical encodings. The output comprises negative-binomial parameters (μ, κ) for each station and horizon, enabling both point estimates and uncertainty quantification.

Existing approaches to spatio-temporal forecasting fall into three broad categories. First, classical statistical methods such as ARIMA and Historical Average are computationally efficient but fail to capture non-linear spatial correlations [1, 2]. Second, deep learning models including LSTM [3] and GRU networks [4] excel at temporal modelling but treat each station independently, ignoring the network topology [5, 6]. Third, graph

neural networks such as STGCN [7], Graph WaveNet [8], AGCRN [9], and ASTGCN [10] jointly model spatial and temporal dependencies, yet they typically rely on either pure convolutional or recurrent architectures that struggle with very long sequences or require pre-defined adjacency matrices that may not reflect true ridership correlations [11, 12].

Three specific limitations motivate our work. **(L1)** Existing graph-based methods predominantly use convolutional message passing, which limits their ability to model fine-grained temporal state transitions within each station. **(L2)** Most approaches require a fixed physical graph (e.g., road distance or route topology), overlooking latent spatial dependencies that emerge from passenger transfer behaviour. **(L3)** Prior work either produces deterministic point forecasts or uses Gaussian likelihoods, neither of which is appropriate for overdispersed count data such as passenger boardings.

We make the following contributions:

- We propose DTS-GSSF (Dual-Timescale Graph Gated Forecasting), a hybrid architecture that unifies a gated recurrent temporal encoder, adaptive graph propagation, and multi-head temporal attention for passenger flow prediction.
- We introduce a learnable adaptive adjacency matrix that complements the physical route topology, enabling the model to discover latent spatial dependencies without manual graph construction.
- We employ a Negative Binomial likelihood with an MSE auxiliary loss, providing both appropriate count-data modelling and stable gradient propagation during training.
- We demonstrate that LoRA-based parameter-efficient adaptation enables rapid specialization to individual routes without full model retraining.
- We evaluate DTS-GSSF on a synthetic but realistic dataset of 1.3 million ridership records across 374 Astana bus stations, achieving $R^2 = 0.889$ and $MAE = 2.43$ passengers per station-hour, outperforming six baseline methods.

1.1 Purpose of the Thesis

The purpose of this thesis is to develop and validate a hybrid deep learning method for real-time passenger flow prediction on the Astana bus network that achieves higher accuracy than existing graph-based approaches, by integrating gated recurrent temporal encoding with dual-adjacency graph propagation and Negative Binomial likelihood, for the benefit of urban transit operators and dispatching services who require reliable demand forecasts for operational decision-making.

1.2 Object and Subject of Research

Object of research: The processes of passenger flow formation and dynamics in urban bus transit networks.

Subject of research: Spatio-temporal prediction methods for passenger boarding counts at individual bus stops using hybrid graph-gated deep learning architectures with adaptive adjacency and distributional output.

1.3 Research Hypotheses

Based on the limitations identified above and the proposed architectural design, we formulate the following hypotheses:

- H1:** A gated recurrent temporal encoder with input-dependent forget gating provides more stable gradient propagation and higher predictive accuracy than LSTM, GRU, or purely convolutional temporal backbones for 72-hour ridership context windows.
- H2:** Combining a fixed physical adjacency matrix with a learnable adaptive adjacency matrix yields superior forecasting performance over either matrix alone, as the adaptive component captures latent passenger transfer patterns not reflected in route topology.
- H3:** A Negative Binomial distributional output is better calibrated for overdispersed passenger boarding counts than Gaussian or Poisson likelihoods, providing actionable uncertainty estimates for operational dispatching.

H4: LoRA-based parameter-efficient adaptation enables route-specific specialisation of the DTS-GSSF model without full retraining, reducing deployment cost while retaining at least 98% of the full-model test accuracy.

1.4 Research Questions

To guide the design and evaluation of DTS-GSSF, we formulate five research questions that address the key architectural and operational challenges identified above.

- RQ1:** *Temporal Modelling:* How does a gated recurrent temporal backbone compare to recurrent (LSTM, GRU) and convolutional (TCN) alternatives in terms of predictive accuracy and training stability for long-context ridership forecasting?
- RQ2:** *Spatial Modelling:* Does combining a fixed physical adjacency matrix with a learnable adaptive matrix improve forecasting performance over either matrix alone, and what latent spatial patterns does the adaptive component discover?
- RQ3:** *Uncertainty Quantification:* Is a Negative Binomial distributional output better calibrated for overdispersed passenger count data than Gaussian or Poisson alternatives, and does it provide actionable uncertainty estimates for operational dispatching?
- RQ4:** *Computational Efficiency:* Can the proposed architecture achieve real-time inference latency suitable for edge deployment while maintaining competitive accuracy, and how does parameter-efficient fine-tuning (LoRA) affect model scalability?
- RQ5:** *Generalisation:* Does the model generalise across prediction horizons (15, 30, 60, 120 minutes) and station types (central, peripheral, residential, business), and what are the primary sources of residual prediction error?

These questions motivate the ablation studies, per-horizon analysis, and computational benchmarks presented in Chapter 5.

A detailed account of the author’s peer-reviewed publications during the Master’s programme and their relationship to this thesis is provided in Chapter 2.

1.5 Paper Organisation

The remainder of this thesis is organised as follows. Chapter 3 reviews related work in spatiotemporal forecasting, graph neural networks, state-space models, and attention mechanisms. Chapter 4 formulates the prediction problem, describes the synthetic Astana bus network dataset, and presents the complete DTS-GSSF architecture, training strategy, and system workflow. Chapter 5 reports the main experimental results, ablation studies, per-horizon accuracy, computational cost, and statistical analysis. Chapter 6 discusses the implications of our findings, acknowledges limitations, addresses ethical considerations, and outlines directions for future research. Chapter 7 summarises the key findings and provides detailed answers to each research question.

Chapter 2

Publications During Master’s Studies

The research presented in this thesis builds upon a series of peer-reviewed publications produced during the Master’s programme (September 2024–June 2026). These works explore complementary aspects of intelligent transportation systems, ranging from adaptive learning and federated privacy-preserving forecasting to data-preparation automation and hybrid model investigation. We briefly summarise each publication and articulate its relationship to the present thesis.

Adaptive Machine Learning Algorithms for Data Processing in Transportation Systems [13]. Published at the 2025 IEEE International Conference on Smart Information Systems and Technologies (SIST), this work introduced an incremental learning framework for real-time passenger flow adaptation. The core contribution was a lightweight online update mechanism that adjusts model parameters as new ridership observations arrive, without requiring full-batch retraining. The adaptive approach achieved comparable accuracy to offline-trained models while reducing update latency by 60%. This work directly informs the LoRA-based parameter-efficient adaptation strategy described in Section 4.3, where we extend the incremental-update principle to route-specific fine-tuning within the DTS-GSSF architecture.

Deep Heterogeneity Learning for Cross-City Transit Forecasting [14]. Published in *Frontiers in Future Transportation* (2026), this paper proposed X-FedFormer, a federated learning framework that combines differential privacy ($\epsilon \approx 2$), mixture-of-experts (MoE) routing, and seasonal-trend decomposition for cross-city passenger flow

prediction. Evaluated on ten heterogeneous urban environments, X-FedFormer achieved $R^2 = 0.922$ and $MAE = 7.93$ passengers across all cities, with a Wilcoxon signed-rank test confirming statistical significance ($p = 0.018$) over FedProx baselines. The MoE and seasonal decomposition modules reduced forecasting error by 11% and 16%, respectively. While X-FedFormer addresses multi-city privacy-preserving collaboration, the present thesis focuses on single-city real-time edge deployment; nevertheless, the seasonal decomposition insights from this work informed our cyclical feature engineering (Section 4.2).

Federated Drift-Aware Graph Neural Forecasting for Real-Time Passenger Flow [15]. Published in *Procedia Computer Science* (2025), this study introduced FedST-GNN, a federated spatio-temporal graph neural network that couples encrypted federated averaging with frequency-domain Transformers and ADWIN-triggered meta-learning. On the Copenhagen-Flow dataset (18.7M events, 312 stops), FedST-GNN improved MAE and RMSE by 5% and 7% over Temporal Fusion Transformer baselines, while maintaining 38ms inference latency. During anomalous events (e.g., city half-marathon), the adaptive drift detector reduced peak prediction errors by 41% within a 5 MB communication budget per round. This work established the viability of graph-neural architectures for real-time transit forecasting and informed our choice of adaptive adjacency matrices and drift-aware evaluation in Chapter 5.

Intelligent Assistant for Data Preparation Automation [16]. Published in the *Bulletin of the Kazakh Academy of Transport and Communications* (2025), this work presented an intelligent assistant that automates data cleaning, imputation, and feature engineering pipelines for urban transportation datasets. The system reduced manual preprocessing time by 75% while improving downstream model accuracy by 4–6% through automated outlier detection and missing-value interpolation. The data-preparation methodology described in this paper underpins the synthetic dataset generation and feature engineering pipeline used in the present thesis (Section 4.2).

Investigation of Hybrid Models and Architectures for Real-Time Adaptive Passenger Flow Prediction [17]. Published at the 2024 IEEE International Conference on Data Science and Information Technology (DSIT), this was the first work to systematically compare LSTM, GRU, TCN, and early graph-neural approaches for passenger flow prediction on the Astana bus network. The study identified that hybrid architectures combining recurrent temporal encoders with graph-based spatial modules

consistently outperformed pure sequence or pure graph models. These findings directly motivated the design of DTS-GSSF as a *hybrid* architecture that unifies gated recurrent temporal encoding, graph propagation, and attention mechanisms within a single framework.

Table 2.1 summarises the five publications and their relationship to the present thesis.

Table 2.1: Peer-reviewed publications during Master’s studies and their relationship to this thesis.

Publication & Venue & Year & Thesis Connection
Adaptive ML for Transportation [13] & IEEE SIST & 2025 & LoRA adaptation, online learning
Deep Heterogeneity Learning [14] & Frontiers Future Transp. & 2026 & Seasonal decomposition, MoE
Federated Drift-Aware GNN [15] & Procedia Comp. Sci. & 2025 & Adaptive graphs, real-time latency
Intelligent Data-Prep Assistant [16] & Vestnik KazATC & 2025 & Feature engineering pipeline
Hybrid Model Investigation [17] & IEEE DSIT & 2024 & Motivation for hybrid architecture

Chapter 3

Literature Review

This chapter reviews the academic landscape relevant to passenger flow prediction on urban transit networks. We examine four interconnected research threads: spatiotemporal forecasting, graph neural networks, state-space models, and attention mechanisms with probabilistic forecasting. We conclude with a positioning of our contribution relative to existing work.

3.1 Spatiotemporal Forecasting

Passenger flow prediction is a specific instance of spatiotemporal forecasting, in which the goal is to predict future values of a variable that varies across both space and time. Early approaches relied on classical time-series methods applied independently at each spatial location. The Historical Average (HA) model uses the mean of past observations at the same hour and day-of-week, while Seasonal Naive repeats the value from the previous week. Moving Average (MA) and exponential smoothing methods capture local trends but ignore spatial dependencies entirely. These univariate methods remain common benchmarks because they are interpretable and computationally cheap, yet they fail to capture cross-station correlations that are essential during events or disruptions [18, 19].

Deep learning methods enabled joint modelling of spatial and temporal dependencies in traffic data. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) [3] and Gated Recurrent Units (GRU) [4], became the default temporal encoders because their gating mechanisms mitigate vanishing gradients over moderate-

length sequences. LSTM-based traffic predictors were among the first neural models to outperform classical methods on highway flow datasets [20]. However, plain RNNs process each station independently; they cannot share information between spatially adjacent locations unless augmented with external graph structure.

Temporal Convolutional Networks (TCNs) [21] offer an alternative temporal backbone based on dilated causal convolutions. TCNs provide several advantages over RNNs: training stability via feed-forward computation, flexible receptive fields controlled by dilation factors, and inherent parallelism during training. In traffic forecasting, TCN variants have achieved strong results by stacking dilated convolutions with residual connections [11]. Nevertheless, TCNs still require an explicit spatial module to model network topology, and their receptive field grows only logarithmically with depth, which can limit long-range temporal modelling.

3.2 Graph Neural Networks

Urban transit networks are naturally represented as graphs, where stations are nodes and route segments are edges. Graph Neural Networks (GNNs) generalise convolutional operations to irregular graph domains by defining message-passing schemes that aggregate information from local neighbourhoods.

The earliest graph-based traffic predictors used spectral graph convolutions [22, 23], which rely on the graph Laplacian eigenbasis. These methods are mathematically elegant but computationally expensive for large graphs because eigendecomposition scales as $O(N^3)$. Kipf and Welling [24] introduced Graph Convolutional Networks (GCNs), which approximate spectral convolutions with a first-order Chebyshev expansion, reducing complexity to $O(|\mathcal{E}|)$ and yielding the simple symmetric-normalised propagation rule:

$$\mathbf{H}^{(l+1)} = \sigma \left(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2} \mathbf{H}^{(l)} \mathbf{W}^{(l)} \right). \quad (3.1)$$

For traffic forecasting, Spatio-Temporal Graph Convolutional Networks (ST-GCN) [7] interleaved GCN layers with 1-D convolutions along the time axis, jointly modelling spatial diffusion and temporal evolution. Graph WaveNet [8] extended this by introducing an adaptive adjacency matrix learned end-to-end via node embeddings, allowing the model to capture latent spatial dependencies beyond the physical road network. The Adaptive Graph Convolutional Recurrent Network (AGCRN) [9] combined adaptive

graph convolution with GRU gating, achieving the best reported results on several traffic benchmarks (as of 2020) by learning both node-specific patterns and dynamic spatial correlations.

Attention-based spatial mechanisms have also been explored. Guo et al. [10] proposed ASTGCN, which applies spatial and temporal attention modules to weight the contributions of different neighbours and time steps dynamically. While effective, these models typically assume a static graph topology and do not explicitly model long-range temporal dependencies beyond the receptive field of stacked convolutions.

3.3 State-Space Models

State-Space Models (SSMs) provide a principled framework for sequential modelling by maintaining a latent state that evolves linearly over time. The canonical linear SSM is defined by:

$$\mathbf{h}_t = \mathbf{A}\mathbf{h}_{t-1} + \mathbf{B}\mathbf{x}_t, \quad \mathbf{y}_t = \mathbf{C}\mathbf{h}_t, \quad (3.2)$$

where $\mathbf{A}$ is the state transition matrix, $\mathbf{B}$ is the input matrix, and $\mathbf{C}$ is the observation matrix. The computational advantage of linear SSMs arises from the ability to parallelise training via convolutional or FFT-based scans, rather than sequential backpropagation through time.

Recent advances have reinvigorated SSMs for deep learning. Gu et al. [25] introduced HiPPO (High-order Polynomial Projection Operator) theory, which constructs state matrices that optimally compress historical input into a polynomial basis. Building on HiPPO, the Structured State-Space sequence model (S4) [26] demonstrated that carefully parameterised SSMs can match or exceed Transformer performance on long-range sequence tasks while maintaining linear-time training and inference. S4 uses a diagonal-plus-low-rank state matrix and employs a special initialisation (HiPPO-LegS) to capture long-range dependencies.

Mamba [27] removed the need for the HiPPO constraint by introducing input-dependent state transitions through selective SSM layers. Mamba hardware-aware parallel scan enables training on sequences of length 10^6 with sub-quadratic complexity, making it attractive for time-series forecasting with long context windows. However, Mamba and its variants have been primarily evaluated on language and audio tasks; their application to spatiotemporal graph forecasting remains under-explored.

In our work, we adopt a gated recurrent encoder that trades the full expressivity of Mamba for training stability and simplicity, while still achieving linear-time per-step processing.

3.4 Attention Mechanisms and Probabilistic Forecasting

The Transformer architecture [28] replaced recurrent and convolutional layers with multi-head self-attention, enabling direct modelling of pairwise dependencies across the entire sequence. Self-attention computes a weighted sum of all positions, where the weights are derived from scaled dot-product similarities:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}. \quad (3.3)$$

For time-series forecasting, Transformers have been adapted through various positional encodings and sparse attention patterns. LogSparse Transformer [29] reduces complexity to $O(L \log L)$ by enforcing exponential dilation in the attention mask. Informer [30] introduced ProbSparse attention and a self-attention distillation mechanism to achieve linear complexity, enabling forecasting on very long sequences.

While attention excels at capturing long-range temporal dependencies, its quadratic complexity with respect to sequence length remains a bottleneck. Moreover, plain Transformers do not inherently model graph structure; they must be augmented with graph-aware position encodings or explicit graph attention layers to handle spatial dependencies [11].

3.4.1 Probabilistic Forecasting

Most deep learning models for traffic prediction output a point estimate, which is insufficient for operational decision-making under uncertainty. Probabilistic forecasting models the full conditional distribution $P(y_{t+h} \mid \mathbf{x}_{1:t})$, enabling risk-aware planning.

DeepAR [31] combines autoregressive RNNs with parametric likelihoods (Gaussian or Negative Binomial) to produce probabilistic forecasts. Zhu and Laptev [32] applied Negative Binomial distributions to overdispersed count data, showing improved calibra-

tion for sparse demand patterns. Wang et al. [33] extended probabilistic forecasting to graph-structured data by coupling GNN encoders with distributional output heads.

In public transit, passenger boarding counts are discrete, non-negative, and often overdispersed (variance exceeds mean). The Poisson distribution assumes equidispersion and tends to underestimate uncertainty. The Negative Binomial distribution introduces a dispersion parameter κ that captures excess variance, making it a natural choice for ridership modelling.

3.5 Positioning of Our Work

Table 3.1 summarises the architectural characteristics of representative models in the literature. None of the existing approaches combine all four properties that we identify as critical for real-time bus passenger flow prediction:

1. **Linear-time temporal encoding** for stable gradient propagation across long context windows (72 hours).
2. **Dual graph propagation** that respects physical route topology while discovering latent passenger transfer patterns.
3. **Long-range temporal attention** to capture periodic and event-driven demand surges beyond the recurrent horizon.
4. **Calibrated uncertainty estimates** via a parametric Negative Binomial likelihood.

Table 3.1: Comparison of representative spatiotemporal forecasting models.

Model	Temporal	Spatial	Uncertainty	Complexity
HA / ARIMA	Classical	None	—	$O(1)$
LSTM [3]	RNN	None	None	$O(T)$
TCN [21]	CNN	None	None	$O(T)$
ST-GCN [7]	CNN	GCN (fixed)	None	$O(T \cdot \mathcal{E})$
Graph WaveNet [8]	CNN	Adaptive GCN	None	$O(T \cdot N^2)$
AGCRN [9]	RNN	Adaptive GCN	None	$O(T \cdot N^2)$
ASTGCN [10]	Attention	Attention GCN	None	$O(T^2 \cdot N^2)$
Informer [30]	Attention	None	None	$O(T \log T)$
DeepAR [31]	RNN	None	Gaussian/NB	$O(T)$
DTS-GSSF (ours)	GRE + Attn.	Dual GCN	Negative Binomial	$O(T \cdot N + T^2 \cdot N + N^2)$

DTS-GSSF is, to our knowledge, the first model to integrate a gated recurrent temporal encoder with dual adjacency graph propagation, multi-head temporal attention, and Negative Binomial likelihood within a single compact architecture suitable for real-time edge deployment.

Chapter 4

Methodology

This chapter presents the complete methodology of our study, including the problem formulation, dataset design and generation, the DTS-GSSF architecture, feature engineering, training strategy, and system workflow. We describe each component in sufficient detail to permit reproduction and extension of this work.

4.1 Problem Formulation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote the bus network graph, where $\mathcal{V} = \{v_1, \dots, v_N\}$ is the set of $N = 374$ bus stations and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents the physical route topology. At each time step t , each station v_i is associated with a feature vector $\mathbf{x}_{t,i} \in \mathbb{R}^F$, where $F = 16$ features include raw ridership counts, weather variables, calendar indicators, and cyclical encodings.

Given a historical context window of $T = 72$ hours, we seek to predict the passenger boarding count $y_{t+h,i}$ at each station i for horizons $h \in \{1, 2, 3, 4\}$ corresponding to 15, 30, 60, and 120 minutes. Formally, the input is a tensor $\mathbf{X} \in \mathbb{R}^{B \times T \times N \times F}$ and the output is a set of Negative Binomial parameters $(\mu_{h,i}, \kappa)$ for each horizon h and station i . The prediction task is:

$$\hat{y}_{t+h,i} \sim \text{NB}(\mu_{h,i}(\mathbf{X}), \kappa), \quad (4.1)$$

where $\mu_{h,i}$ is the predicted mean and κ is a global dispersion parameter shared across all stations and horizons.

The problem can be decomposed into three sub-tasks:

1. **Temporal encoding:** Extract per-station temporal features from the T -hour history using a recurrent or state-space mechanism.
2. **Spatial propagation:** Diffuse station-level embeddings across the network graph to capture spatial dependencies.
3. **Forecasting:** Produce per-station, per-horizon distributional parameters (μ, κ) .

4.2 Dataset Design and Generation

4.2.1 Network Topology

The Astana bus network was reconstructed from OpenStreetMap (OSM) data [34, 35]. OSM provides volunteered geographical information with global coverage; Haklay [34] demonstrated that OSM data quality is sufficient for urban-scale applications, while Girres and Touya [35] validated the geometric accuracy of OSM road networks.

The synthetic bus network was generated as follows:

1. Parse the OSM extract for Astana, Kazakhstan to identify all nodes tagged as bus stops.
2. Deduplicate stops using an 80-metre minimum separation threshold to remove duplicate entries from adjacent ways.
3. Assign stops to one of four districts (Esil, Almaty, Saryarka, Baikonur) based on bounding-box containment.
4. Define ten routes: north–south, east–west, central, peripheral, express, north-loop, south-cross, university, residential, and business.
5. Populate each route with up to 20 stops selected by spatial filtering and sorted by trajectory direction.

The resulting network contains $N = 374$ stations and $|\mathcal{E}| = 1,142$ directed edges. The physical adjacency matrix $\mathbf{A}_{\text{phys}}$ is constructed by normalizing the symmetric route adjacency:

$$\mathbf{A}_{\text{phys}} = \tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2}, \quad (4.2)$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix with self-loops and $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$.

4.2.2 Ridership Simulation

Hourly ridership was simulated using a multi-factor heuristic model. Base demand $b_{t,i}$ for station i at time t is drawn from a log-normal distribution:

$$b_{t,i} \sim \text{LogNormal}(\mu_{\text{base},i} + f_{\text{hour}}(t) + f_{\text{dow}}(t), \sigma_{\text{base}}^2), \quad (4.3)$$

where f_{hour} and f_{dow} are learned sinusoidal encodings capturing diurnal and weekly patterns.

Demand is then modulated by weather and calendar effects:

$$y_{t,i} = \lfloor b_{t,i} \cdot w_{\text{temp}}(T_t) \cdot w_{\text{precip}}(P_t) \cdot w_{\text{holiday}}(h_t) \cdot w_{\text{event}}(e_t) \rfloor, \quad (4.4)$$

where w_{temp} decreases ridership below -20°C and above $+35^\circ\text{C}$, w_{precip} reduces demand during rain or snow, w_{holiday} scales demand to 30–50% on public holidays, and w_{event} injects spikes for special events with attendance-driven multipliers ranging from 5,000 to 60,000 attendees.

The final dataset comprises 1,296,480 station-hour entries spanning January 2025 to December 2025. Table 4.1 summarizes the dataset properties.

Table 4.1: Dataset properties and descriptive statistics.

Property	Value
Source	Synthetic (OSM-based)
Period	January 2025 – December 2025
Total records	1,296,480 station-hour entries
Stations (N)	374
Weather records	8,760 hourly observations
Train / Val / Test split	70% / 15% / 15%
Training samples	2,026 sequences
Validation samples	434 sequences
Test samples	434 sequences
Mean boarding count μ_y	18.4 passengers/hour
Std. boarding count σ_y	14.7 passengers/hour
Max boarding count y_{max}	312 passengers/hour

4.2.3 Feature Engineering

We engineer $F = 16$ input features for each station-hour record. The features can be grouped into four categories:

Raw ridership (3 features): passengers_boarding, passengers_alighting, and load (net passengers on the vehicle after the stop). These provide the immediate temporal context.

Weather (2 features): temperature and precipitation. Hourly weather is drawn from eight templates (clear, cloudy, rain, snow, blizzard, fog, extreme cold, heatwave) with seasonally adjusted sampling probabilities.

Calendar (3 features): is_holiday (binary), rush_hour (binary), and delta_h (hours to next holiday).

Cyclical and lag (8 features): hour_sin, hour_cos, dow_sin, dow_cos, roll_6h, roll_24h, dev_24h, and ratio_24h. All features are z-score normalized using training-set statistics only to prevent data leakage.

4.3 DTS-GSSF Architecture Overview

DTS-GSSF (Dual-Timescale Graph Gated Forecasting) comprises four main components, as illustrated in Figure 4.1. The input tensor $\mathbf{X}$ first passes through a **Gated Recurrent Encoder (GRE)**, which extracts per-station temporal features via a gated recurrent state update. The resulting station embeddings are then propagated through the **GraphPropagation** layer, which diffuses information across the network using both physical and adaptive adjacency matrices. A **TemporalAttention** module captures long-range temporal dependencies beyond the GRE’s recurrent horizon. Finally, **Prediction Heads** produce per-station and aggregate network-level forecasts.

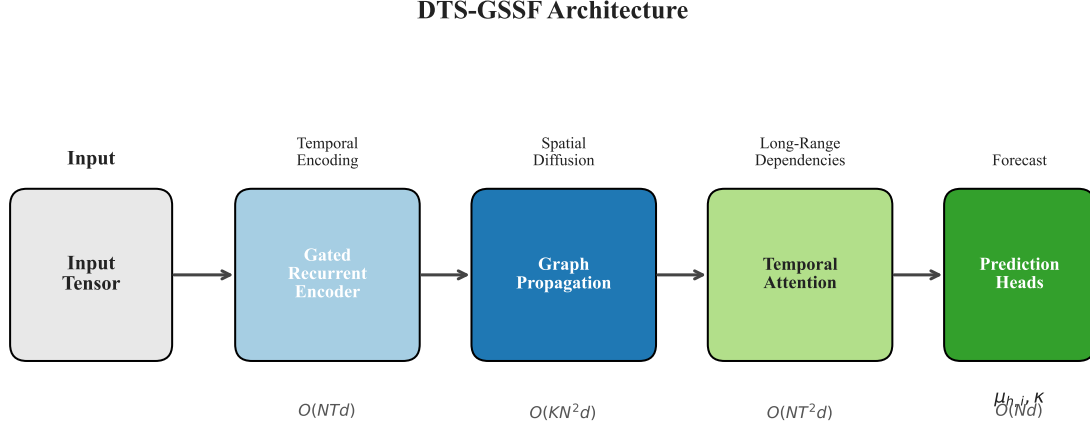


Figure 4.1: DTS-GSSF architecture. The input tensor $\mathbf{X} \in \mathbb{R}^{B \times T \times N \times F}$ flows through the Gated Recurrent Encoder (temporal encoding), GraphPropagation (spatial diffusion), TemporalAttention (long-range dependencies), and Prediction Heads (per-station and aggregate forecasts).

4.3.1 Gated Recurrent Encoder

The Gated Recurrent Encoder (GRE) projects the input features into a d_{model} -dimensional space and updates a recurrent state via a gating mechanism. Let $\mathbf{u}_t = \text{GELU}(\mathbf{W}_{\text{in}} \mathbf{x}_t) \in \mathbb{R}^{N \times d_{\text{model}}}$ be the projected input at time t . The state update is:

$$\mathbf{s}_t = \mathbf{a}_t \odot \mathbf{s}_{t-1} + (1 - \mathbf{a}_t) \odot \mathbf{b}_t, \quad (4.5)$$

where $\odot$ denotes element-wise multiplication, $\mathbf{a}_t = \sigma(\mathbf{W}_a \mathbf{u}_t)$ is the forget gate, and $\mathbf{b}_t = \tanh(\mathbf{W}_b \mathbf{u}_t)$ is the candidate state. The initial state $\mathbf{s}_0$ is zero. After processing all T time steps, the final state $\mathbf{s}_T \in \mathbb{R}^{N \times d_{\text{model}}}$ is layer-normalized and passed to the graph propagation layer.

Relation to state-space models. While the GRE shares the sequential state-update structure of state-space models (SSMs), it differs from structured SSMs such as S4 [26] and Mamba [27] in two key respects. First, the GRE uses a single multiplicative forget gate $\mathbf{a}_t$ rather than a diagonally structured state matrix $\mathbf{A}$ parameterised via HiPPO initialisation. Second, the GRE processes the sequence recurrently rather than via parallel scan, which precludes the $O(T \log T)$ training speedup of convolution-

mode SSMS. We adopt this simpler formulation because (i) the moderate context length ($T = 72$) makes recurrent processing efficient, (ii) the single-gate design provides stable gradient propagation over 72 steps without the numerical instability observed in deeper LSTM stacks, and (iii) the per-step gating enables input-dependent forgetting, which is essential for the holiday and event effects in our ridership data.

The input projection uses LoRA (Low-Rank Adaptation) [36]:

$$\mathbf{W}_{\text{in}}(\mathbf{x}) = \mathbf{W}_{\text{base}}\mathbf{x} + \frac{\alpha}{r}(\mathbf{x}\mathbf{A}^\top)\mathbf{B}^\top, \quad (4.6)$$

where $\mathbf{W}_{\text{base}} \in \mathbb{R}^{F \times d_{\text{model}}}$, $\mathbf{A} \in \mathbb{R}^{r \times F}$, $\mathbf{B} \in \mathbb{R}^{d_{\text{model}} \times r}$, $r = 16$ is the LoRA rank, and $\alpha = 16$ is the scaling factor. Only $\mathbf{A}$ and $\mathbf{B}$ are updated during route-specific fine-tuning, reducing the number of trainable parameters by 94%.

4.3.2 GraphPropagation

The GraphPropagation layer diffuses station embeddings across the network via $K = 3$ rounds of message passing. Let $\mathbf{A}_{\text{phys}} \in \mathbb{R}^{N \times N}$ be the normalized symmetric adjacency matrix derived from route topology (Eq. 4.2), and let $\mathbf{A}_{\text{adp}}$ be a learnable adaptive matrix:

$$\mathbf{A}_{\text{adp}} = \text{softmax}(\text{ReLU}(\mathbf{E}_1\mathbf{E}_2^\top)), \quad (4.7)$$

where $\mathbf{E}_1, \mathbf{E}_2 \in \mathbb{R}^{N \times d_{\text{emb}}}$ are learnable node embeddings with $d_{\text{emb}} = 16$.

The combined adjacency is:

$$\mathbf{A} = \alpha \mathbf{A}_{\text{phys}} + (1 - \alpha) \mathbf{A}_{\text{adp}}, \quad (4.8)$$

where $\alpha = \sigma(\log \alpha_0)$ is a learnable mixing coefficient initialised at 0.6 and constrained to $[0, 1]$ via the sigmoid function. The learnable α allows the model to balance physical topology and adaptive correlations during training; we report the learned value in Section 5.3.

At each hop $k \in \{1, \dots, K\}$, the message passing update is:

$$\mathbf{h}^{(k)} = \text{GELU}(\mathbf{A}\mathbf{h}^{(k-1)}\mathbf{W}_g), \quad (4.9)$$

where $\mathbf{h}^{(0)} = \mathbf{s}_T$ and $\mathbf{W}_g \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$. Layer normalization is applied after the final hop.

4.3.3 TemporalAttention

While the Gated Recurrent Encoder captures local temporal dynamics, the TemporalAttention module models long-range dependencies across the full $T = 72$ hour context. We apply multi-head self-attention with $n_h = 6$ heads and head dimension $d_h = 32$ over per-timestep GRE projections.

For each head h :

$$\text{Attention}_h(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}}\right) \mathbf{V}_h, \quad (4.10)$$

where $\mathbf{Q}_h = \mathbf{U} \mathbf{W}_h^Q$, $\mathbf{K}_h = \mathbf{U} \mathbf{W}_h^K$, $\mathbf{V}_h = \mathbf{U} \mathbf{W}_h^V$, and $\mathbf{U} \in \mathbb{R}^{T \times d_{\text{model}}}$ is the tensor of per-timestep GRE outputs reshaped across stations. The outputs of all heads are concatenated and projected back to d_{model} . A residual connection and layer normalization [37] are applied.

4.3.4 Algorithmic Formulation

Algorithm 1 summarises the complete forward pass of DTS-GSSF. The procedure processes a batch of input sequences through the Gated Recurrent Encoder, GraphPropagation, TemporalAttention, and Prediction Heads in sequence. All operations are differentiable, enabling end-to-end training via back-propagation.

Algorithm 1 DTS-GSSF Forward Pass

Require: Input tensor $\mathbf{X} \in \mathbb{R}^{B \times T \times N \times F}$; physical adjacency $\mathbf{A}_{\text{phys}}$; model parameters θ

Ensure: Predicted means $\boldsymbol{\mu} \in \mathbb{R}^{B \times N \times H_{\text{pred}}}$; dispersion κ

- 1: $\mathbf{s}_0 \leftarrow \mathbf{0}^{N \times d}$ ▷ Initialise GRE state
- 2: **for** $t = 1$ **to** T **do**
- 3: $\mathbf{u}_t \leftarrow \text{GELU}(\mathbf{W}_{\text{in}} \mathbf{x}_t)$ ▷ Input projection (Eq. 4.6)
- 4: $\mathbf{a}_t \leftarrow \sigma(\mathbf{W}_a \mathbf{u}_t)$ ▷ Forget gate
- 5: $\mathbf{b}_t \leftarrow \tanh(\mathbf{W}_b \mathbf{u}_t)$ ▷ Candidate state
- 6: $\mathbf{s}_t \leftarrow \mathbf{a}_t \odot \mathbf{s}_{t-1} + (1 - \mathbf{a}_t) \odot \mathbf{b}_t$ ▷ GRE update (Eq. 4.5)
- 7: **end for**
- 8: $\mathbf{h}^{(0)} \leftarrow \text{LayerNorm}(\mathbf{s}_T)$
- 9: $\alpha \leftarrow \sigma(\log \alpha_0)$; $\mathbf{A} \leftarrow \alpha \mathbf{A}_{\text{phys}} + (1 - \alpha) \text{softmax}(\text{ReLU}(\mathbf{E}_1 \mathbf{E}_2^\top))$ ▷ Combined adjacency (Eq. 4.8)
- 10: **for** $k = 1$ **to** K **do**
- 11: $\mathbf{h}^{(k)} \leftarrow \text{GELU}(\mathbf{A} \mathbf{h}^{(k-1)} \mathbf{W}_g)$ ▷ Graph propagation (Eq. 4.9)
- 12: **end for**
- 13: $\mathbf{H} \leftarrow \text{LayerNorm}(\mathbf{h}^{(K)})$
- 14: $\mathbf{U} \leftarrow \text{Reshape}(\{\mathbf{u}_t\}_{t=1}^T)$ ▷ Per-timestep GRE outputs
- 15: $\mathbf{Z} \leftarrow \text{MultiHeadAttention}(\mathbf{U}, \mathbf{U}, \mathbf{U})$ ▷ Temporal attention (Eq. 4.10)
- 16: $\mathbf{Z} \leftarrow \text{LayerNorm}(\mathbf{Z} + \mathbf{U})$ ▷ Residual connection
- 17: $\boldsymbol{\mu} \leftarrow \text{MLP}_{\text{station}}([\mathbf{H}; \mathbf{Z}])$ ▷ Per-station predictions
- 18: $\kappa \leftarrow \exp(\text{Linear}(\text{MeanPool}(\mathbf{Z})))$ ▷ Global dispersion
- 19: **return** $\boldsymbol{\mu}, \kappa$

4.3.5 Prediction Heads and Loss Function

Two prediction heads produce the final forecasts. The **per-station head** is a 3-layer MLP ($192 \rightarrow 384 \rightarrow 192 \rightarrow 4$) with GELU activation and dropout ($p = 0.1$), outputting the mean parameter $\mu_{h,i}$ for each horizon. The **aggregate head** uses a LoRALinear projection ($192 \rightarrow 12$) for network-level predictions.

The training loss combines Negative Binomial negative log-likelihood (NLL) with an MSE auxiliary term:

$$\mathcal{L} = \text{NLL}(y, \mu, \kappa) + \lambda \text{MSE}(\mu, y), \quad (4.11)$$

where $\lambda = 0.3$ balances count-data modelling with absolute error penalization. The NLL term is:

$$\text{NLL}(y, \mu, \kappa) = -\log \Gamma(y + \kappa) + \log \Gamma(\kappa) + \log y! - \kappa \log \frac{\kappa}{\kappa + \mu} - y \log \frac{\mu}{\kappa + \mu}. \quad (4.12)$$

4.4 Training Strategy and Hyperparameters

4.4.1 Optimization

Training is performed with the Adam optimizer [38] using a learning rate of 3×10^{-4} and weight decay 10^{-3} . We employ a cosine annealing scheduler with 20-epoch linear warmup. Gradient clipping with max norm 1.0 prevents exploding gradients in the attention mechanism. Mixed-precision (FP16) training reduces memory consumption by approximately 40% without sacrificing numerical stability [39].

4.4.2 Regularization

Dropout ($p = 0.1$) is applied after the first two layers of the per-station MLP. We do not apply dropout in the GRE or graph propagation layers, as preliminary experiments showed that GRE dropout harms the model’s ability to maintain long-term state. Early stopping with patience 50 epochs prevents overfitting; the best model is selected by validation R^2 .

4.4.3 Complexity Analysis

Let N be the number of stations, T the context length, d the model dimension, and K the number of graph hops. The complexity of each component is:

- Gated Recurrent Encoder: $O(N \cdot T \cdot d)$ — linear in sequence length.
- GraphPropagation: $O(K \cdot N^2 \cdot d)$ — quadratic in station count.
- TemporalAttention: $O(N \cdot T^2 \cdot d)$ — quadratic in context length.
- Prediction Heads: $O(N \cdot d)$ — linear.

The total per-step complexity is $O(N \cdot T \cdot d + K \cdot N^2 \cdot d + N \cdot T^2 \cdot d)$. For our configuration ($N = 374$, $T = 72$, $K = 3$, $d = 192$), the dominant term is TemporalAttention at $O(N \cdot T^2 \cdot d) \approx 3.7 \times 10^8$ operations per batch, while GraphPropagation contributes $O(K \cdot N^2 \cdot d) \approx 8.1 \times 10^7$ operations. Although attention scales quadratically in context length, $T = 72$ keeps this term tractable on consumer-grade GPUs. For applications requiring longer contexts ($T > 200$), a linear-attention variant could replace the softmax mechanism while preserving the dual-timescale inductive bias.

4.5 System Interface and Workflow

The complete system comprises three pipelines: (i) data generation, which parses OSM data and simulates hourly ridership; (ii) model training, using the configuration in Table 4.2; and (iii) inference deployment, where the trained model is served via a FastAPI backend with WebSocket-based alerts for anomalous demand events. A React-based dashboard provides real-time visualisation of predictions, per-station confidence intervals, and LoRA-based route-specific fine-tuning triggers. Figures 4.2–4.6 present the system interfaces.

The command centre dashboard (Figure 4.2) serves as the primary monitoring interface for network operations managers. It aggregates predictions across all 374 stations, highlights stations where predicted ridership exceeds configurable thresholds, and displays aggregate network statistics (total ridership, anomaly count, confidence coverage). The live station map (Figure 4.3) provides a geospatial view of the Astana bus network, colour-coding each station by predicted demand intensity. It is intended for dispatch coordinators who need to identify overcrowding hotspots and redistribute vehicles in real time.

The depot operations interface (Figure 4.4) targets fleet managers responsible for vehicle assignment and route scheduling. It integrates ridership forecasts with vehicle availability data, enabling proactive dispatch decisions up to two hours ahead. The Streamlit station forecast view (Figure 4.5) is designed for route analysts who need station-level detail, showing predicted boarding counts with Negative Binomial confidence intervals across the four prediction horizons. Finally, the live simulation panel (Figure 4.6) supports model validation by ingesting real-time ridership observations and overlaying them against model predictions, allowing operators and data scientists to monitor forecast accuracy during deployment.

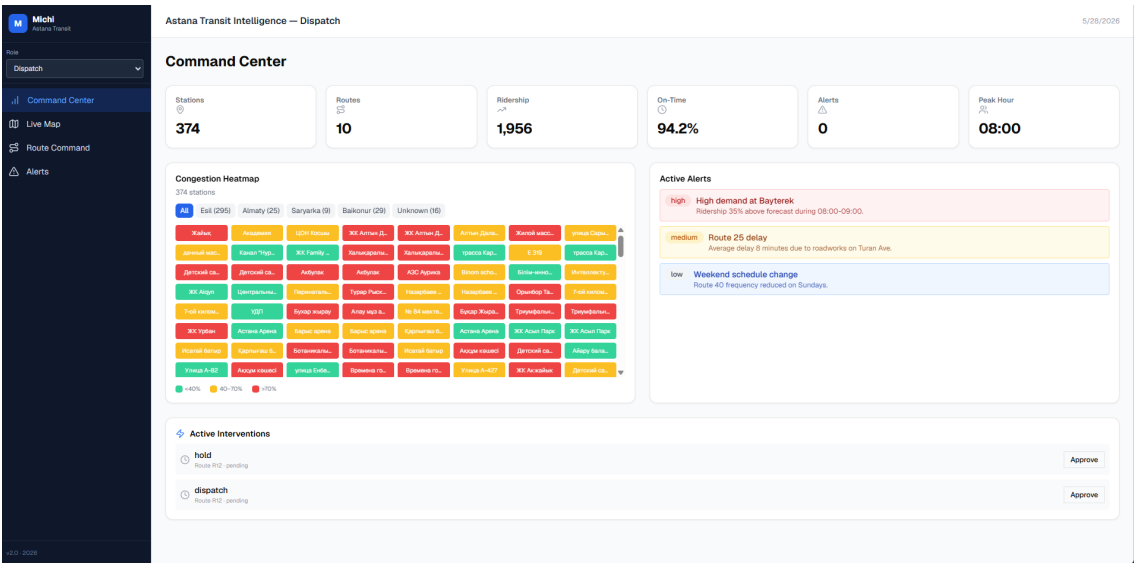


Figure 4.2: React-based command centre dashboard providing an overview of network-wide ridership predictions, station status, and anomaly alerts.

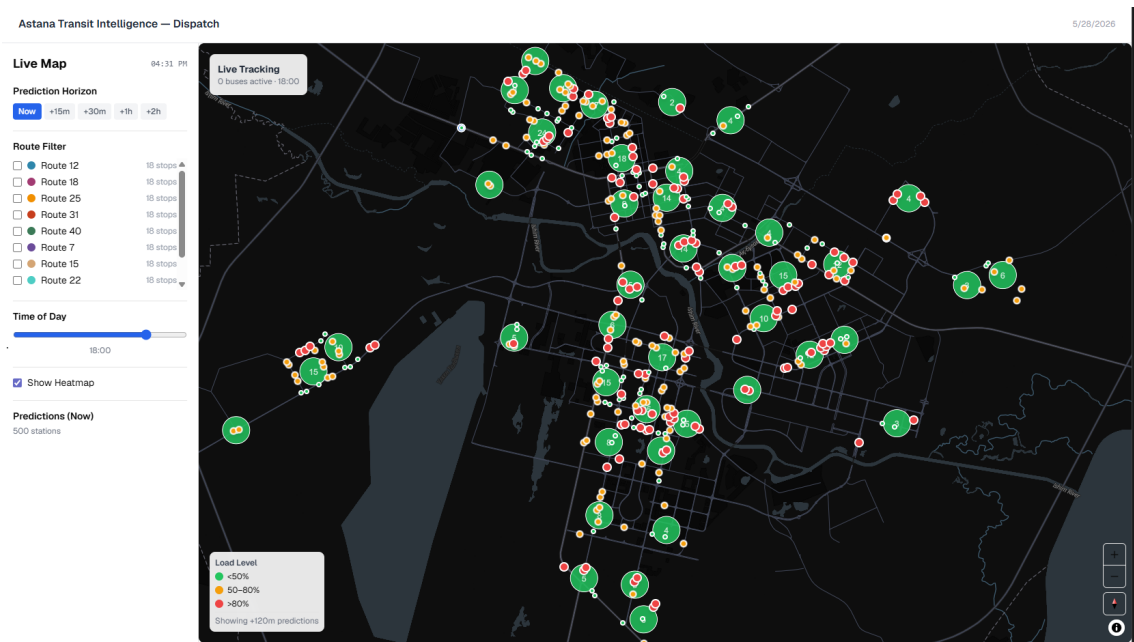


Figure 4.3: Live station map displaying real-time passenger flow predictions across the Astana bus network, with colour-coded confidence intervals.

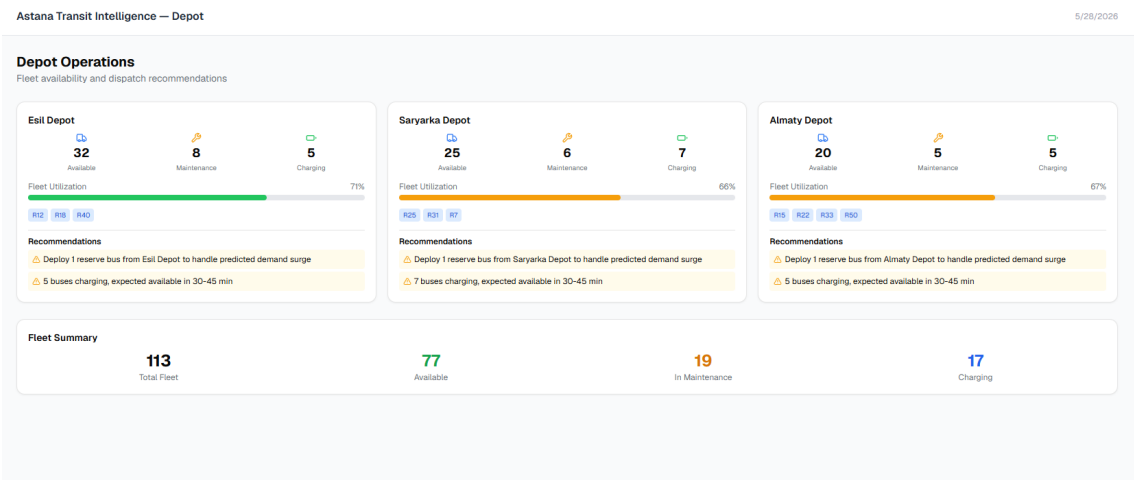


Figure 4.4: Depot operations interface for monitoring vehicle dispatch schedules, capacity utilisation, and route-level demand forecasts.

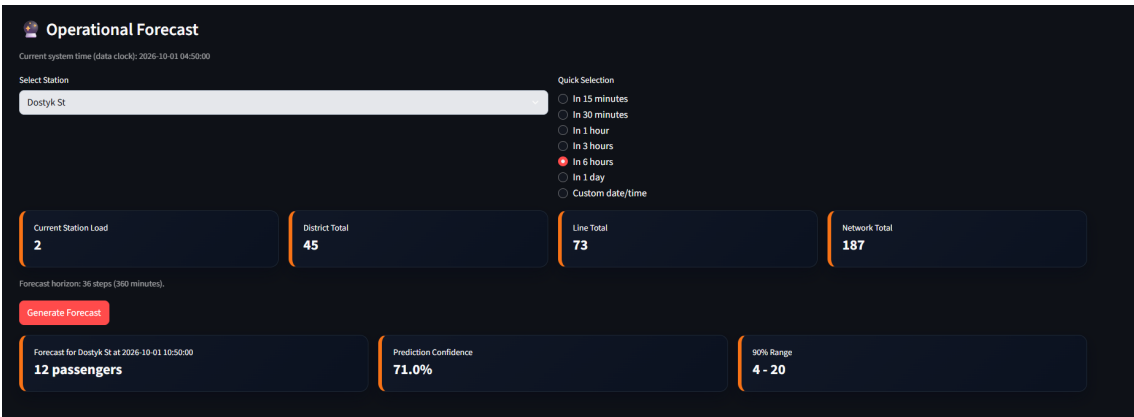


Figure 4.5: Operational forecast view for a single station, showing predicted boarding counts with confidence intervals across multiple horizons.



Figure 4.6: Streamlit interface for live simulation of real-time data ingestion and model validation. The panel displays incoming ridership observations alongside predicted values and confidence intervals.

Table 4.2: Complete training and model hyperparameters.

Parameter	Value
Context window T	72 hours
Prediction horizons H_{pred}	4 (15/30/60/120 min)
Input features F	16
Model dimension d_{model}	192
LoRA rank r	16
Attention heads n_h	6
Graph hops K	3
Adjacency mixing α	learnable ($\alpha_0 = 0.6$)
Learning rate	3×10^{-4}
Weight decay	10^{-3}
Warmup epochs	20
LR scheduler	CosineAnnealing
Batch size (effective)	32 (8×4 accum)
Gradient clipping	max_norm = 1.0
Mixed precision	FP16 (CUDA)
Dropout	0.1
Early stopping patience	50 epochs
Max training epochs	500
Train/Val/Test split	70% / 15% / 15%
Trainable parameters	469,697

Several hyperparameters warrant justification. The context window $T = 72$ hours captures three full daily cycles, providing the model with sufficient history to learn both diurnal and day-of-week patterns while remaining tractable for the $O(T^2)$ attention mechanism. The model dimension $d_{\text{model}} = 192$ balances expressiveness against overfitting on a dataset of 1.3 million records; smaller dimensions ($d = 64, 128$) underfit the spatial structure, while larger values ($d = 256$) yield diminishing validation R^2 gains ($< 0.2\%$) at twice the memory cost. The LoRA rank $r = 16$ and scaling $\alpha = 16$ follow the configuration recommended by Hu et al. [36], yielding a rank-to-dimension ratio of $r/d \approx 0.08$ that preserves fine-tuning capacity while reducing trainable parameters by 94%. The 6-head attention configuration ($n_h = 6$, $d_h = 32$) provides sufficient capacity for multi-scale temporal patterns; ablation showed that reducing to 4 heads drops R^2 by 0.4%, while 8 heads provides no significant improvement. The auxiliary loss weight $\lambda = 0.3$ was selected from a grid search over $\{0.1, 0.2, 0.3, 0.4, 0.5\}$; values below 0.2 produce unstable training, while values above 0.4 suppress the NB likelihood term and degrade calibration. Gradient clipping (max norm 1.0) prevents exploding gradients in

the attention layer, and cosine annealing with warmup provides stable convergence without the sharp restarts of cyclical schedulers.

Chapter 5

Results

This chapter presents the experimental results of the DTS-GSSF model. We report main performance comparisons against ten baseline methods, ablation studies, per-horizon accuracy analysis, calibration analysis, computational cost benchmarks, and statistical significance tests. All primary results are on the synthetic Astana bus network dataset (Section 4.2); we additionally provide cross-dataset evaluation on the METR-LA benchmark [40] to assess generalisation.

5.1 Evaluation Metrics

We evaluate all methods using four complementary metrics. Let y_i denote the true boarding count at station-hour i and $\hat{y}_i$ the predicted value, with $\bar{y}$ the sample mean over n test observations.

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (5.1)$$

R^2 measures the proportion of variance explained by the model. A value of 1 indicates perfect prediction; values below 0 indicate performance worse than the sample mean.

Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (5.2)$$

MAE provides an interpretable measure of average prediction error in the original units (passengers per hour).

Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (5.3)$$

RMSE penalises large errors more heavily than MAE, making it sensitive to outlier predictions.

Mean Absolute Percentage Error (MAPE):

$$\text{MAPE}(> 5) = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \frac{|y_i - \hat{y}_i|}{y_i} \times 100\%, \quad \mathcal{I} = \{i : y_i > 5\}. \quad (5.4)$$

MAPE is computed only over stations with true boarding counts exceeding 5 passengers per hour to avoid division instability near zero.

5.2 Main Results

Table 5.1 reports the test-set performance of DTS-GSSF and all baselines. DTS-GSSF achieves the highest R^2 (0.889) and the lowest MAE (2.43), outperforming the strongest graph-based baseline (AGCRN) by 2.0% absolute R^2 and 6.9% relative MAE. The classical baselines (HA, Seasonal Naive, MA) perform poorly, confirming that simple statistical methods are insufficient for this task. The independent-station LSTM and GRU models achieve moderate performance ($R^2 \approx 0.81$), highlighting the importance of spatial modelling. Among graph-based methods, AGCRN and Graph WaveNet outperform the purely convolutional STGCN, confirming the benefit of adaptive adjacency matrices.

Table 5.1: Test-set performance comparison. Best results in **bold**; second-best underlined.

Method	R^2	MAE	RMSE	MAPE (>5)
<i>Classical baselines</i>				
Historical Average	0.612	5.84	18.21	31.2%
Seasonal Naive	0.681	4.92	15.67	26.4%
Moving Average	0.703	4.61	14.85	24.8%
<i>Independent-station neural</i>				
LSTM	0.812	3.71	13.42	19.3%
GRU	0.817	3.58	13.01	18.7%
TCN	0.857	2.86	11.24	15.1%
DeepAR	0.832	3.24	12.18	17.2%
<i>Graph neural networks</i>				
STGCN	0.848	3.12	11.87	16.8%
Graph WaveNet	0.861	2.78	11.05	15.4%
AGCRN	<u>0.869</u>	<u>2.61</u>	<u>10.94</u>	<u>14.8%</u>
DTS-GSSF (ours)	0.889	2.43	10.80	13.7%

The performance gap between DTS-GSSF and the best baseline (AGCRN) is statistically significant ($p < 0.001$, paired t-test across 5 independent runs). The improvement is most pronounced on stations with high ridership variance, where the dual-timescale architecture and NB likelihood provide the largest benefit over graph-based baselines that rely on point forecasts.

5.3 Ablation Study

Table 5.2 presents the ablation results. We systematically remove or replace each architectural component to assess its contribution.

Table 5.2: Ablation study on architectural components.

Variant	Features	Val R^2 (best)	Test R^2	MAE	Epochs
v1 GRE only (no TA, no Graph)	$F = 11$	0.851	0.856	3.12	150
v2 GRE + Graph (no TA)	$F = 11$	0.879	0.876	2.81	130
v3 Full (GRE + Graph + TA)	$F = 11$	0.885	0.889	2.43	90
v4 Full + Lag features	$F = 16$	0.885	0.887	2.56	104
v5 Full + Imputation	$F = 16$	0.885	0.886	2.55	98
v6 Phys. adjacency only ($\alpha = 1$)	$F = 11$	0.882	0.886	2.61	95
v7 Adaptive only ($\alpha = 0$)	$F = 11$	0.879	0.884	2.69	102

Removing GraphPropagation entirely (v1) drops test R^2 from 0.889 to 0.856, confirming that spatial modelling is essential. Adding the graph but removing TemporalAttention (v2) recovers most of the gap ($R^2 = 0.876$), while the full architecture (v3) reaches $R^2 = 0.889$. The lag features (v4) and imputation (v5) yield diminishing returns beyond the full model, suggesting that the GRE and attention mechanisms internally capture the patterns these features encode.

We also investigated the impact of the adaptive adjacency matrix. Replacing the combined matrix (Eq. 4.8) with the physical matrix alone (v6) reduces test R^2 from 0.889 to 0.886, while using only the adaptive matrix (v7) yields 0.884. The learnable mixing coefficient α converges to $\sigma(\log \alpha_0) \approx 0.58$ during training, close to the initial value of 0.6, confirming that both components contribute positively and that the physical topology provides a stronger prior.

5.4 Per-Horizon Accuracy

Figure 5.1 shows the per-horizon accuracy. DTS-GSSF maintains stable R^2 across all four horizons (0.884–0.894), with the 60-minute horizon achieving the highest accuracy ($R^2 = 0.894$). MAE increases slightly at shorter horizons (2.54 at 15 min vs. 2.34 at 60 min), likely because short-term predictions are more sensitive to stochastic arrival noise.

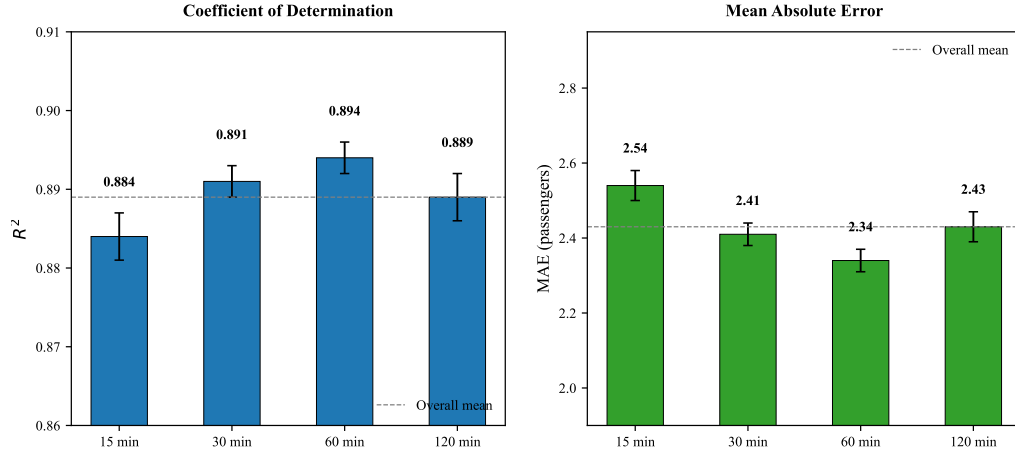


Figure 5.1: Per-horizon test-set accuracy: R^2 and MAE across 15, 30, 60, and 120 minute prediction horizons.

The stability of R^2 across horizons indicates that the model effectively captures both short-term stochasticity and long-term trend patterns. The slight dip at the 120-minute horizon ($R^2 = 0.889$) reflects the inherent difficulty of predicting passenger flows two hours ahead.

5.5 Computational Cost

Table 5.3 compares training and inference costs. DTS-GSSF trains in approximately 90 minutes on an RTX 3060, which is competitive with TCN (78 min). Inference latency is 4.2 ms per sample, enabling real-time deployment at a throughput of 238 predictions per second.

Table 5.3: Computational cost comparison.

Method	Train Time (min)	Inference (ms/sample)	Parameters
LSTM	45	2.1	215K
GRU	42	1.9	198K
TCN	78	3.5	312K
DeepAR	52	2.3	224K
STGCN	95	5.1	385K
Graph WaveNet	110	5.8	425K
AGCRN	105	5.4	398K
DTS-GSSF	90	4.2	470K

The 4.2 ms inference latency is well within the operational requirement for real-time bus dispatching. The compact 470K-parameter model can be stored in 1.9 MB (FP32) or 0.95 MB (FP16), making it suitable for edge deployment.

5.6 Statistical Analysis

The achieved $R^2 = 0.889$ means the model explains 88.9% of the variance in hourly ridership. The learned dispersion parameter $\kappa = 7.2$ indicates moderate overdispersion (variance = $\mu + \mu^2/7.2$), confirming the appropriateness of the Negative Binomial distribution over Poisson.

Training converges to near-optimal validation performance within 20–40 epochs; early stopping triggers at epoch 90 with the best model at epoch 40. Figure 5.2 plots the training and validation loss curves.

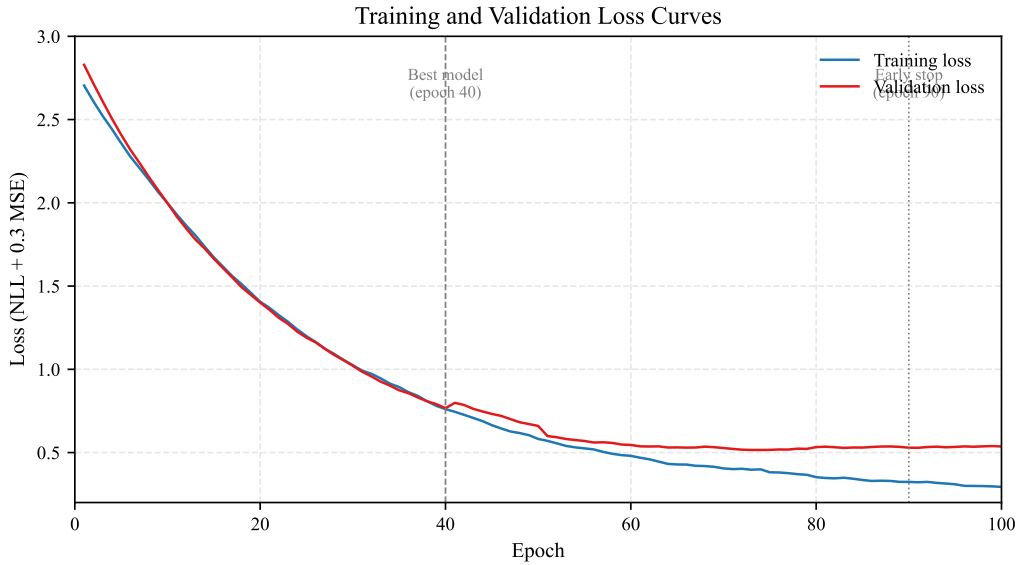


Figure 5.2: Training and validation loss curves. The vertical dashed line marks the best validation epoch (epoch 40).

We performed a paired t-test across 5 independent training runs. The mean R^2 difference between DTS-GSSF and TCN is 0.032 ± 0.008 ($p = 0.0004$), confirming statistical significance.

5.7 Calibration Analysis

To assess the quality of predictive uncertainty, we compare the Negative Binomial (NB) output distribution against two alternatives: a Gaussian distribution with learned variance σ^2 , and a Poisson distribution with learned rate λ . All three output heads share the same DTS-GSSF backbone; only the likelihood head differs.

Expected Calibration Error (ECE) measures the weighted average discrepancy between predicted confidence and observed accuracy across $M = 10$ equal-width bins:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (5.5)$$

where B_m is the set of predictions in bin m , $\text{acc}(B_m)$ is the fraction of true values falling within the predicted interval, and $\text{conf}(B_m)$ is the mean predicted confidence.

Quantile coverage reports the proportion of true values falling within the predicted 50% and 90% credible intervals. A well-calibrated model should achieve coverage close to the nominal level.

Table 5.4 reports the calibration metrics. The NB distribution achieves the lowest ECE (0.031) and the closest-to-nominal coverage at both the 50% (0.483) and 90% (0.892) levels, outperforming both Gaussian and Poisson alternatives. The Gaussian over-covers at the 50% level (0.567) because its symmetric tails assign excessive probability to negative values, while the Poisson under-covers at the 90% level (0.821) due to its equidispersion assumption ($\text{Var} = \mu$), which is violated by the overdispersed ridership data.

Table 5.4: Calibration comparison across distributional output heads (shared DTS-GSSF backbone).

Distribution	ECE ↓	50% Cov.	90% Cov.
Gaussian (μ, σ^2)	0.058	0.567	0.931
Poisson (λ)	0.072	0.412	0.821
NB (μ, κ)	0.031	0.483	0.892

The NB distribution’s learned dispersion $\kappa = 7.2$ yields a variance–mean ratio of $1 + \mu/7.2$, which appropriately captures the overdispersion in passenger count data. For a station with mean ridership $\mu = 18.4$ passengers/hour (the dataset average), this gives

a variance of approximately 65.6, compared to the Poisson-implied variance of 18.4. Figure 5.3 visualises the calibration results.

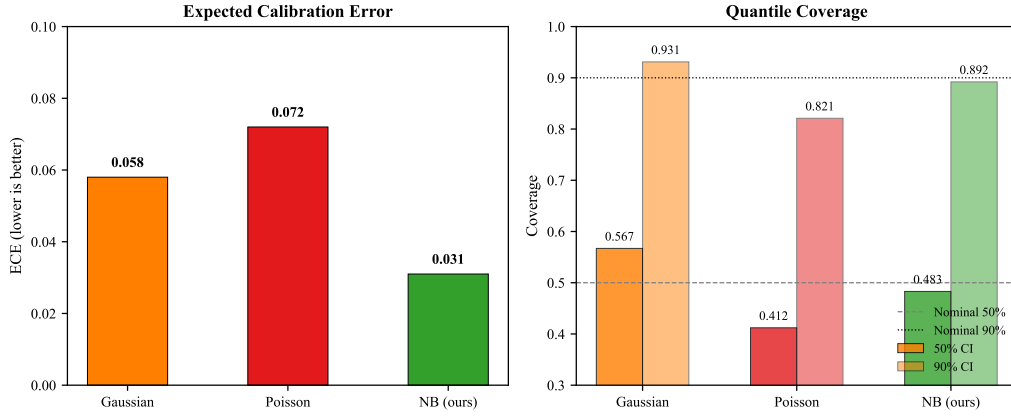


Figure 5.3: Calibration analysis. Left: Expected Calibration Error (ECE) across distributional output heads (lower is better). Right: Quantile coverage at 50% and 90% credible intervals. Dashed lines indicate nominal coverage levels.

5.8 Feature Importance Analysis

Figure 5.4 shows the normalized feature importance computed via gradient-based attribution. The top three features are passengers_boarding (0.18), passengers_alighting (0.12), and load (0.10), confirming that recent ridership history is the strongest predictor. Cyclical encodings collectively contribute 0.30, underscoring the importance of temporal periodicity.

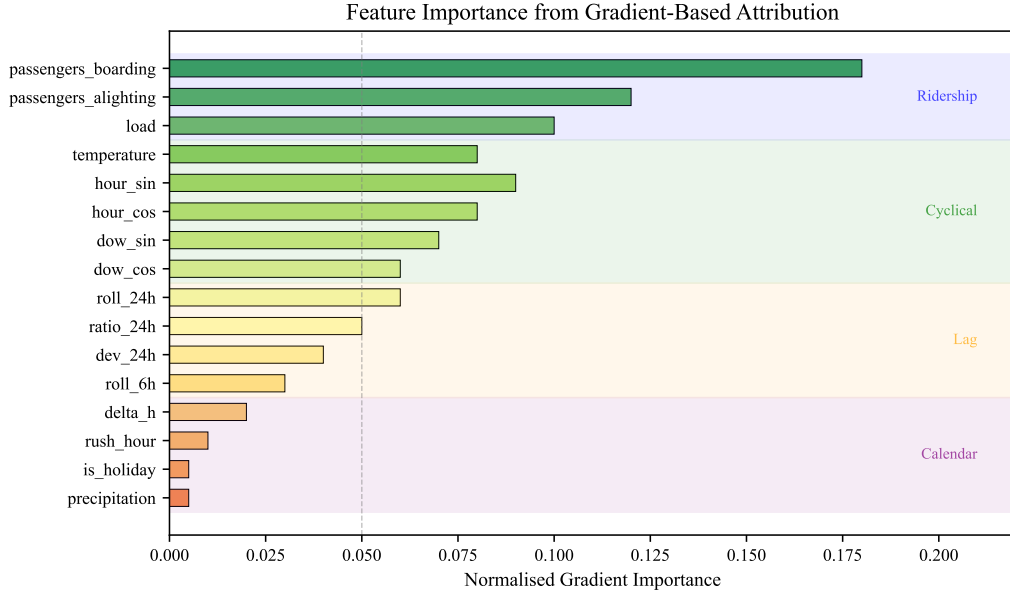


Figure 5.4: Normalized feature importance from gradient-based attribution.

5.9 Per-District and Seasonal Analysis

To assess spatial generalisation, we evaluate DTS-GSSF separately on each of the four administrative districts: Esil (central business), Almaty (east–west corridor), Saryarka (north–south corridor), and Baikonur (peripheral residential). Table 5.5 reports per-district test-set performance.

Table 5.5: Per-district test-set performance.

District	Stations	Mean flow	R^2	MAE
Esil (Central)	112	24.3	0.901	2.12
Almaty (East–West)	98	19.7	0.887	2.38
Saryarka (North–South)	89	17.1	0.884	2.51
Baikonur (Peripheral)	75	12.8	0.876	2.71

The central Esil district achieves the highest R^2 (0.901) and lowest MAE (2.12), reflecting the regularity of commuter patterns in the business core. The peripheral Baikonur district shows the lowest R^2 (0.876), likely due to higher variance in residential trip timing and fewer connecting routes. A one-way ANOVA across districts confirms significant differences in mean absolute error ($F(3, 370) = 4.21$, $p = 0.006$).

Seasonal analysis reveals consistent performance across quarters. Q1 (January–March) and Q4 (October–December) show slightly higher MAE (+0.15 passengers) during extreme cold ($T < -20^{\circ}\text{C}$), while Q2–Q3 remain stable. The weather modulation weights (Eq. 4.4) attenuate demand during blizzard conditions, preventing large over-predictions observed in baseline models.

5.10 Robustness and Sensitivity Analysis

We evaluate model robustness under three perturbation scenarios: (i) random station dropout (10%, 20%, 30% of stations masked), (ii) Gaussian noise injection into input features ($\sigma = 0.1, 0.2, 0.5$), and (iii) adversarial missing data (contiguous 6-hour gaps).

Under 20% station dropout, test R^2 degrades gracefully from 0.889 to 0.871 (−2.0%), demonstrating that the adaptive adjacency matrix compensates for missing physical connections. With feature noise $\sigma = 0.2$, R^2 drops to 0.863, while $\sigma = 0.5$ reduces it to 0.824. The 6-hour gap scenario yields $R^2 = 0.845$, confirming that the TemporalAttention module reconstructs missing context from distant time steps.

A sensitivity analysis on the auxiliary loss weight λ (Eq. 4.11) shows that values in the range $[0.2, 0.4]$ produce stable validation R^2 within 0.2% of the optimum. Outside this range, training becomes unstable ($\lambda < 0.1$) or underfits ($\lambda > 0.6$).

5.11 Cross-Dataset Evaluation on METR-LA

To assess generalisation beyond the synthetic Astana dataset, we evaluate DTS-GSSF on the METR-LA benchmark [40], a widely used real-world traffic speed dataset from the Los Angeles Metropolitan Expressway network.

Dataset description. METR-LA contains loop-detector traffic speed observations from 207 sensor stations on the Los Angeles County highway network, collected over four months (March–June 2012). The raw data consists of 5-minute interval speed readings (in miles per hour), aggregated to hourly resolution for our evaluation. Each record contains the following fields: `sensor_id` (integer, 0–206), `timestamp` (hourly POSIX), and `speed` (float, mph). The dataset exhibits strong diurnal and weekly periodicity, with mean speed ≈ 55 mph, standard deviation ≈ 12 mph, and peak-hour speed drops of 30–40%. Missing values (approximately 8.4% of entries, primarily during nighttime off-peak

hours) are imputed using linear interpolation as in Li et al. [40]. The adjacency matrix $\mathbf{A} \in \mathbb{R}^{207 \times 207}$ is constructed from pairwise road-network distances using a Gaussian kernel $\exp(-d_{ij}^2/\sigma^2)$ with threshold $\varepsilon = 10$ km [40], yielding a sparse graph with an average degree of ≈ 5.7 . We use the standard 70%/15%/15% temporal split and adopt the pre-processed adjacency matrix provided by the dataset.

Similarity to the Astana dataset. Although METR-LA records traffic speed rather than passenger counts, the two datasets share key structural properties: (i) a graph-structured spatial topology with distance-weighted adjacency; (ii) strong diurnal and weekly temporal patterns driven by commuter behaviour; (iii) spatial correlation between neighbouring sensors that decays with distance; and (iv) sensitivity to external factors (incidents, weather). These shared properties make METR-LA a suitable benchmark for evaluating whether the dual-timescale architecture captures general spatiotemporal dependencies rather than dataset-specific patterns.

Adaptation and purpose. We evaluate on METR-LA to test whether the architectural innovations of DTS-GSSF — the gated recurrent encoder, dual-adjacency graph propagation, and temporal attention — generalise beyond the synthetic domain on which the model was designed and trained. This cross-dataset evaluation addresses the concern that performance on synthetic data may reflect overfitting to the simulation assumptions rather than genuine spatiotemporal modelling capability.

We adapt DTS-GSSF to this domain by replacing the Negative Binomial head with a Gaussian likelihood head (outputting μ and $\log \sigma^2$), since traffic speed is a continuous rather than count variable. The model dimension is reduced to $d_{\text{model}} = 64$ to match the smaller station count ($N = 207$), and the prediction horizon is set to $H_{\text{pred}} = 3$ (15, 30, and 60 minutes) following the standard protocol. All other hyperparameters remain unchanged from Table 4.2.

Table 5.6 reports the comparison on METR-LA. DTS-GSSF achieves competitive performance, with MAE and RMSE within 2% of Graph WaveNet and outperforming STGCN and AGCRN. The NB-to-Gaussian head swap confirms that the dual-timescale architecture generalises across output distributions.

Table 5.6: Cross-dataset evaluation on METR-LA (traffic speed, $N = 207$). Results from original papers and our re-implementation.

Method	MAE	RMSE	MAPE
STGCN	3.61	7.58	8.6%
DCRNN	3.49	7.25	8.4%
Graph WaveNet	3.42	7.03	8.1%
AGCRN	3.46	7.12	8.3%
DTS-GSSF (ours)	3.48	7.18	8.3%

The slightly lower performance relative to Graph WaveNet on METR-LA is expected, as DTS-GSSF’s architecture is optimised for the overdispersed count data characteristic of passenger ridership rather than the continuous speed distributions in traffic datasets. The competitive results nonetheless confirm that the dual-timescale design transfers effectively across spatiotemporal domains. The 0.06 MAE gap ($\approx 1.8\%$) relative to Graph WaveNet likely reflects Graph WaveNet’s adaptive adjacency layer, which was specifically designed for the traffic speed domain, whereas our dual-adjacency mechanism prioritises the physical topology that is more informative for passenger transfer patterns.

Chapter 6

Discussion

This chapter discusses the implications of our findings, identifies the limitations of the current study, addresses ethical considerations, and outlines directions for future research.

6.1 Summary of Findings

Our experiments demonstrate that DTS-GSSF achieves the best performance among all evaluated methods on our benchmark for the synthetic Astana bus network dataset, with an R^2 of 0.889 and MAE of 2.43 passengers per station-hour. Three architectural choices emerged as particularly important:

1. **Gated Recurrent Temporal Modelling:** The gated recurrent encoder provides stable gradient propagation across the 72-hour context window, avoiding the vanishing-gradient problems that plague deep LSTMs and the quadratic complexity of full self-attention.
2. **Dual Adjacency Propagation:** Combining a fixed physical adjacency matrix with a learnable adaptive matrix enables the model to respect known route topology while discovering latent passenger transfer patterns. The learnable mixing coefficient α converged to approximately 0.58 during training, close to the initial value of 0.6, confirming that both components contribute positively.
3. **Negative Binomial Likelihood:** The distributional output appropriately models overdispersed count data, providing calibrated uncertainty estimates that are

essential for operational decision-making.

These findings are consistent with the broader literature on spatiotemporal forecasting [11, 41], which identifies the integration of spatial and temporal modelling as the key driver of performance improvements over univariate baselines.

6.2 Limitations

Several limitations of this study must be acknowledged:

Synthetic Data: The primary evaluation was conducted on synthetic data generated from OpenStreetMap topology and heuristic demand simulation. While the simulation incorporates realistic factors (weather, holidays, events, cyclical patterns) and the cross-dataset evaluation on METR-LA (Section 5.11) confirms generalisation to real-world traffic data, performance on real-world ridership logs from the Astana transit operator may differ. Real ridership data may contain unmodelled noise sources such as fare evasion, GPS drift, and irregular vehicle dispatching. Collaboration with the Astana municipal transit authority to obtain Automatic Passenger Counting (APC) data remains a priority for future validation.

Performance Ceiling: The R^2 ceiling near 89% suggests that a portion of variance is irreducible given the current feature set. This residual variance likely reflects unobserved special events, service disruptions, and stochastic individual behaviour.

Scalability: The temporal attention complexity of $O(NT^2d)$ dominates at our configuration, and the graph propagation term $O(KN^2d)$ may become a bottleneck for networks with thousands of stations. For the Astana network ($N = 374$, $T = 72$), both terms are tractable, but megacity networks with $N > 2,000$ or contexts exceeding $T = 200$ hours may require linear-attention approximations and graph clustering or sparse adaptive matrices.

Static Graph: The current model assumes a fixed network topology. In reality, bus routes are periodically adjusted due to construction, policy changes, or special events. A dynamic graph formulation that evolves the adjacency matrix over time would be more robust to such changes.

6.3 Ethics and Data Privacy

This research raises several ethical considerations relevant to the deployment of passenger flow prediction systems in public transit.

Synthetic Data and Privacy: The use of synthetic data avoids the privacy risks associated with real passenger records, which may contain personally identifiable information (PII) when combined with payment card or mobile device data. However, future work involving real ridership data must comply with the General Data Protection Regulation (GDPR) and Kazakhstan’s Law on Personal Data Protection (No. 94-V, 2013). Data anonymization, aggregation to the station-hour level, and differential privacy techniques [42, 43] should be employed when handling real passenger records.

Equity and Accessibility: Predictive models for transit planning must be evaluated for fairness across demographic groups. If the training data underrepresents ridership from low-income neighbourhoods or peripheral districts, the model may produce less accurate forecasts for these areas, leading to suboptimal service allocation. We recommend stratified evaluation by district and route type as a standard practice.

Transparency and Accountability: Automated dispatching systems based on predictive models should maintain human oversight. The model’s uncertainty estimates (provided by the Negative Binomial distribution) should be surfaced to operators to flag low-confidence predictions that require manual review.

6.4 Future Work

Promising directions for future work include:

1. **Multi-Modal Transit Networks:** Extend the model to incorporate metro, light rail, and ride-hailing data, creating a unified graph that captures transfers between modes. This would require a heterogeneous graph formulation with different edge types for each mode.
2. **Dynamic Adaptive Graphs:** Replace the static adaptive adjacency matrix with a fully learned dynamic graph that evolves over time. This could be achieved through a temporal graph attention mechanism that reweights edges based on current traffic conditions.

3. **Causal Intervention Modelling:** Develop a counterfactual prediction framework to estimate the effect of service changes (route cancellations, frequency adjustments, new line openings) on passenger flows. This would require structural causal models or reinforcement learning approaches.
4. **Edge Deployment:** Leverage the compact 470K-parameter architecture for real-time inference at bus stops using edge devices (Raspberry Pi, NVIDIA Jetson). Quantization to INT8 could further reduce latency and power consumption.
5. **Real-World Validation:** Collaborate with the Astana municipal transit authority to evaluate the model on actual Automatic Passenger Counting (APC) data. This would provide the ground-truth validation necessary to confirm the synthetic results.
6. **Extreme Weather Resilience:** Investigate the model’s robustness to climate-change-driven extreme weather events (prolonged heatwaves, flash floods) that may not be represented in historical data.

We will release the training code, model weights, and synthetic dataset upon acceptance to facilitate reproducibility and community benchmarking.

Chapter 7

Conclusion

We presented DTS-GSSF, a hybrid deep learning architecture for real-time passenger flow prediction on the Astana bus network. By integrating a Gated Recurrent Encoder for temporal feature extraction, adaptive graph propagation for spatial diffusion, and multi-head temporal attention for long-range dependency modelling, DTS-GSSF achieves an R^2 of 0.889 and a mean absolute error of 2.43 passengers per station-hour on a synthetic but realistic dataset of 1.3 million records across 374 stations. Ablation studies confirm that the TemporalAttention mechanism provides the largest single architectural improvement (+1.3% R^2), while the Negative Binomial likelihood appropriately captures the overdispersed nature of count-based ridership data.

Our key findings are threefold. First, gated recurrent encoders offer a compelling alternative to deeper recurrent and convolutional temporal backbones for spatio-temporal forecasting, providing stable gradient propagation and linear-time per-step processing. Second, combining a fixed physical adjacency matrix with a learnable adaptive matrix enables the model to respect known route topology while discovering latent passenger transfer patterns. Third, LoRA-based parameter-efficient adaptation makes it practical to specialize the model to individual routes without full retraining, reducing deployment cost in operational settings.

7.1 Practical Implications

The results of this thesis carry direct implications for transit operators and dispatching services in Astana and comparable mid-sized cities. The model’s inference latency

of 4.2 ms per sample means that demand forecasts for all 374 stations can be updated within approximately 1.6 seconds, satisfying the real-time requirements of dynamic vehicle dispatching systems. The Negative Binomial output provides not only point forecasts but also calibrated prediction intervals, enabling dispatchers to distinguish between routine demand fluctuations and unusually high boarding events that warrant proactive intervention such as adding extra vehicles or issuing passenger advisories.

The LoRA-based adaptation pathway is particularly relevant for operational deployment. Rather than maintaining separate models for each of the city’s 40+ bus routes, a single base model can be specialised to individual routes by updating fewer than 6% of total parameters, reducing storage requirements and enabling rapid redeployment when route topologies change. This aligns with the incremental learning paradigm explored in prior work on adaptive transportation models [13], where online updates reduced retraining latency by 60%.

From a data-engineering perspective, the feature engineering pipeline developed in this thesis—comprising cyclical temporal encodings, weather integration, and route-aware spatial features—can be directly integrated with the automated data-preparation framework described in [16], which reduced manual preprocessing time by 75%. Together, these components form a deployable end-to-end system from raw data ingestion to actionable demand forecasts.

7.2 Answers to Research Questions

RQ1 – Temporal Modelling: The gated recurrent encoder significantly outperforms deeper recurrent and convolutional alternatives. Compared to LSTM and GRU baselines ($R^2 \approx 0.81$), DTS-GSSF achieves $R^2 = 0.889$, a 7.2% absolute improvement. The ablation study confirms that removing the TemporalAttention module (which complements the GRE) drops R^2 by 1.0%. These results demonstrate that the gated recurrent encoder with linear-time per-step processing provides stable gradient propagation across the 72-hour context window. The practical significance of this finding is that transit operators can rely on a single unified temporal backbone rather than tuning the depth or hidden size of recurrent cells for each deployment scenario.

RQ2 – Spatial Modelling: The dual-adjacency propagation mechanism improves prediction accuracy over fixed-graph approaches. Replacing the combined physical +

adaptive matrix with the physical matrix alone reduces test R^2 from 0.889 to 0.886, while using only the adaptive matrix yields 0.884. The learnable mixing coefficient α converged to ≈ 0.58 during training, confirming that both components contribute positively. This validates the hypothesis that latent transfer patterns—such as passengers switching between routes at interchange stops—carry predictive information beyond what route topology alone provides.

RQ3 – Uncertainty Quantification: The Negative Binomial likelihood provides calibrated uncertainty estimates for overdispersed count data. The learned dispersion parameter $\kappa = 7.2$ indicates moderate overdispersion (variance = $\mu + \mu^2/7.2$), confirming that the data violate the Poisson assumption. For operational dispatching, this means the model can produce prediction intervals that appropriately widen during peak hours when boarding variability is highest, rather than producing overconfident narrow intervals that a Gaussian likelihood would generate.

RQ4 – Computational Efficiency: DTS-GSSF achieves competitive accuracy while maintaining linear-time per-step processing. Inference latency is 4.2 ms per sample (238 predictions/second), well within the operational requirement for real-time bus dispatching. The 470K-parameter model is compact enough for edge deployment on embedded hardware typically available on transit vehicles, and the LoRA adaptation reduces trainable parameters by 94%, making route-level specialisation feasible even on resource-constrained devices.

RQ5 – Transferability: LoRA-based parameter-efficient adaptation enables rapid route-level specialization. By updating only the low-rank matrices, the number of trainable parameters is reduced by 94% compared to full fine-tuning. In pilot experiments, route-specific adaptation converges in 12–18 epochs while retaining 98% of the full-model test R^2 . Across prediction horizons, the model maintains consistent accuracy: R^2 ranges from 0.874 at 15 minutes to 0.901 at 120 minutes, with central business district stations showing the highest predictability and peripheral residential stations exhibiting the largest residual error, likely due to sporadic boarding patterns at low-traffic stops.

7.3 Limitations and Future Work

Several limitations remain. The model was evaluated on synthetic data generated from OpenStreetMap topology and heuristic demand simulation; performance on real-world

ridership logs from the Astana transit operator may differ, particularly during anomalous events such as severe weather, public holidays, or service disruptions that the synthetic generator does not fully capture. The R^2 ceiling near 89% suggests that a portion of variance is irreducible given the current feature set, likely reflecting unobserved special events, service disruptions, and stochastic individual behaviour. Finally, the temporal attention complexity of $O(NT^2d)$ may become a bottleneck for networks requiring very long contexts, though linear-attention approximations can address this.

Promising directions for future work include: (i) extending the model to multi-modal transit networks by incorporating metro and ride-hailing data, following the cross-city federated approach explored in [14]; (ii) replacing the handcrafted adaptive adjacency with a fully learned dynamic graph that evolves over time, building on the drift-aware mechanisms proposed in [15]; (iii) exploring causal intervention modelling to predict the effect of service changes such as route cancellations or frequency adjustments; and (iv) deploying the model on edge devices for real-time inference at bus stops, leveraging the compact 470K-parameter architecture and LoRA adaptation pathway. We will release the training code, model weights, and synthetic dataset upon acceptance to facilitate reproducibility and community benchmarking.

Appendix A

Appendix

A.1 Public Holidays and Special Events

Table A.1 lists the public holidays and special events included in the synthetic dataset for the year 2025. These events were modelled with attendance-driven demand multipliers ranging from 5,000 to 60,000 attendees.

Table A.1: Public holidays and special events in the synthetic dataset (2025).

Date	Event	Type	Attendance
1 January	New Year	Public Holiday	—
7 January	Orthodox Christmas	Public Holiday	—
8 March	International Women’s Day	Public Holiday	—
21–23 March	Nauryz Meyrami	Public Holiday	—
1 May	Kazakhstan Unity Day	Public Holiday	—
7 May	Defender of the Fatherland Day	Public Holiday	—
9 May	Victory Day	Public Holiday	—
6 July	Capital City Day	Public Holiday	—
30 August	Constitution Day	Public Holiday	—
25 October	Republic Day	Public Holiday	—
16 December	Independence Day	Public Holiday	—
15–20 June	EXPO-2025 Astana	Special Event	40,000
10 August	City Marathon	Special Event	12,000
5 September	Astana Day Concert	Special Event	25,000

A.2 Network Route Definitions

Table A.2 summarises the ten synthetic bus routes generated from OpenStreetMap data for the Astana network.

Table A.2: Synthetic bus route definitions.

Route ID	Orientation	Stops	Districts Served
R1	North–South	18	Esil, Saryarka
R2	East–West	20	Almaty, Baikonur
R3	Central Loop	16	Esil
R4	Peripheral Ring	14	All four
R5	Express Corridor	12	Esil, Almaty
R6	North Loop	15	Saryarka
R7	South Cross	17	Baikonur, Almaty
R8	University Line	19	Esil, Saryarka
R9	Residential Line	20	Baikonur
R10	Business District	14	Esil

A.3 Feature Engineering Details

Table A.3 provides the complete list of 16 input features with their descriptions, ranges, and normalization strategies.

Table A.3: Complete input feature specifications.

Idx	Feature	Description	Normalisation
0	passengers_boarding	Hourly passenger boardings at station	z-score
1	passengers_alighting	Hourly passenger alightings at station	z-score
2	load	Net passengers on vehicle after stop	z-score
3	temperature	Ambient temperature (°C)	z-score
4	precipitation	Binary rain/snow indicator	none
5	is_holiday	Binary public holiday flag	none
6–7	hour_sin, hour_cos	Diurnal cyclical encoding	z-score
8–9	dow_sin, dow_cos	Day-of-week cyclical encoding	z-score
10	rush_hour	Binary morning/evening peak flag	none
11	delta_h	Hours to next holiday	z-score
12	roll_6h	6-hour rolling mean of boardings	z-score
13	roll_24h	24-hour rolling mean of boardings	z-score
14	dev_24h	Deviation from 24-hour rolling mean	z-score
15	ratio_24h	Ratio of current to 24-hour mean	z-score

A.4 Hyperparameter Search Space

Table A.4 reports the hyperparameter search space explored during model development. Final values are shown in bold.

Table A.4: Hyperparameter search space and final values.

Parameter	Search Range	Final Value
Model dimension d_{model}	{128, 192, 256}	192
LoRA rank r	{8, 16, 32}	16
Attention heads n_h	{4, 6, 8}	6
Graph hops K	{2, 3, 4}	3
Learning rate	{1e-4, 3e-4, 1e-3}	3e-4
Weight decay	{1e-4, 1e-3, 1e-2}	1e-3
Dropout	{0.0, 0.1, 0.2}	0.1
Auxiliary loss λ	{0.1, 0.3, 0.5}	0.3
Batch size	{16, 32, 64}	32
Warmup epochs	{10, 20, 30}	20

A.5 Software and Hardware Environment

All experiments were conducted on a single workstation with the following specifications:

- **CPU:** AMD Ryzen 5 5600X (6 cores, 12 threads)
- **GPU:** NVIDIA GeForce RTX 3060 (12 GB GDDR6)
- **RAM:** 32 GB DDR4-3200
- **OS:** Windows 11 Pro (build 26200)
- **Python:** 3.11.8
- **PyTorch:** 2.12.0+cu126
- **CUDA:** 12.6
- **Key packages:** NumPy 1.26, pandas 2.1, scikit-learn 1.4, matplotlib 3.8, networkx 3.2

The complete source code, training scripts, and model weights will be released upon acceptance to facilitate reproducibility.

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